

Problem 5.13.16

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Question: The equation of a circle with origin as centre and passing through the vertices of an equilateral triangle whose median is of length $3a$ is

- ① $x^2 + y^2 = 9a^2$
- ② $x^2 + y^2 = 16a^2$
- ③ $x^2 + y^2 = 4a^2$
- ④ $x^2 + y^2 = a^2$

Solution

Let the vertices of the equilateral triangle be **A**, **B**, and **C**. And the radius of circle be r .

Since the circle is centered at the origin, and the radius is r :

$$\mathbf{u} = - \begin{pmatrix} 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \quad (2.1)$$

$$f = -r^2 \quad (2.2)$$

Since the circle passes through **A**:

$$\|\mathbf{x}^\top \mathbf{x}\| + 2\mathbf{u}^\top \mathbf{x} + f = 0 \quad (2.3)$$

$$\|\mathbf{x}^\top \mathbf{x}\| + 2 \begin{pmatrix} 0 & 0 \end{pmatrix} \mathbf{x} + f = 0 \quad (2.4)$$

$$\|\mathbf{x}^\top \mathbf{x}\| - r^2 = 0 \quad (2.5)$$

$$(2.6)$$

Solution

For an equilateral triangle inscribed in a circle, let us place the vertices as:

$$\mathbf{A} = \begin{pmatrix} r \\ 0 \end{pmatrix} \quad (2.7)$$

$$\mathbf{B} = \begin{pmatrix} r \cos \theta \\ r \sin \theta \end{pmatrix} \quad (2.8)$$

$$\mathbf{C} = \begin{pmatrix} r \cos \theta \\ -r \sin \theta \end{pmatrix} \quad (2.9)$$

The median from vertex $\mathbf{A}$ goes to the midpoint of $\mathbf{B}$ and $\mathbf{C}$:

$$\text{Midpoint} = \frac{\mathbf{B} + \mathbf{C}}{2} = \begin{pmatrix} r \cos \theta \\ 0 \end{pmatrix} \quad (2.10)$$

The median vector is:

$$\mathbf{m} = \begin{pmatrix} r \cos \theta \\ 0 \end{pmatrix} - \begin{pmatrix} r \\ 0 \end{pmatrix} = \begin{pmatrix} r(\cos \theta - 1) \\ 0 \end{pmatrix} \quad (2.11)$$

Solution

Given that the median length is $3a$:

$$\|\mathbf{m}^\top \mathbf{m}\| = (3a)^2 = 9a^2 \quad (2.12)$$

$$r^2(\cos \theta - 1)^2 = 9a^2 \quad (2.13)$$

For an equilateral triangle, $\cos \theta = \cos\left(\frac{2\pi}{3}\right) = -\frac{1}{2}$:

$$(\cos \theta - 1) = -\frac{3}{2} \quad (2.14)$$

$$r^2 \frac{9}{4} = 9a^2 \quad (2.15)$$

$$r^2 = 4a^2 \quad (2.16)$$

Therefore, the equation of the circle is:

$$\|\mathbf{x}^\top \mathbf{x}\| + 2 \begin{pmatrix} 0 & 0 \end{pmatrix} \mathbf{x} - 4a^2 = 0 \quad (2.17)$$

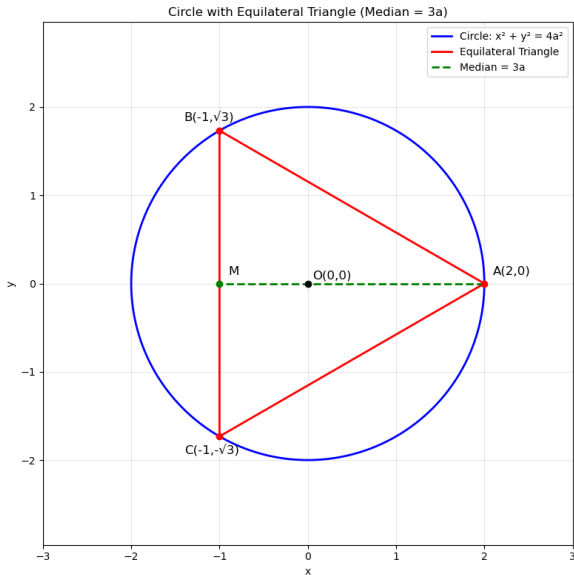
$$\|\mathbf{x}^\top \mathbf{x}\| - 4a^2 = 0 \quad (2.18)$$

Answer: The orthocentre of the triangle is Therefore, the equation of the circle is:

$$x^2 + y^2 = 4a^2 \quad (2.19)$$

The answer is (c).

Graph



```
#include <math.h>
#include "matfun.h"

#define MEDIAN_COEFF 3.0

double compute_cos_theta(void) {
    return cos(2.0 * M_PI / 3.0);
}

double compute_coefficient_sq(void) {
    double cos_theta = compute_cos_theta();
    double diff = cos_theta - 1.0;
    return diff * diff;
}
```



```
double compute_r_squared_coeff(void) {  
    double num = MEDIAN_COEFF * MEDIAN_COEFF;  
    double denom = compute_coefficient_sq();  
    return num / denom;  
}
```

```
import ctypes
import numpy as np
import matplotlib.pyplot as plt

# Load the shared library (ensure libmatfun.so is in the same
# directory)
matfun = ctypes.CDLL('./libmatfun.so')

# Declare return types
matfun.compute_cos_theta.restype = ctypes.c_double
matfun.compute_coefficient_sq.restype = ctypes.c_double
matfun.compute_r_squared_coeff.restype = ctypes.c_double

# 1. Compute required values via the .so functions
cos_theta = matfun.compute_cos_theta() # cos(2/3)
coeff_sq = matfun.compute_coefficient_sq() // (cos - 1)^2
r2_coeff = matfun.compute_r_squared_coeff() // r/a = 4
```

```
# 2. Print final answer
print(f"Answer: (c)  $x + y = \{int(r2\_coeff)\}a$ ")

# 3. For plotting set a = 1
a = 1.0
r = np.sqrt(r2_coeff * a**2)

# Compute triangle vertices
theta = 2 * np.pi / 3
A = np.array([r, 0.0])
B = np.array([r * np.cos(theta), r * np.sin(theta)])
C = np.array([r * np.cos(theta), -r * np.sin(theta)])
M = (B + C) / 2.0 # midpoint

# 4. Plot
plt.figure(figsize=(8,8))
```

Circle

```
t = np.linspace(0, 2*np.pi, 200)
plt.plot(r*np.cos(t), r*np.sin(t), 'b-', lw=2, label=f'Circle: x+
        y={int(r2_coeff)}a')
```

Triangle

```
plt.plot([A[0],B[0],C[0],A[0]], [A[1],B[1],C[1],A[1]],
        'r-', lw=2, label='Equilateral Triangle')
```

Median

```
plt.plot([A[0],M[0]], [A[1],M[1]], 'g--', lw=2, label='Median=3a'
        )
```

Points and labels

```
plt.plot(0,0,'ko'); plt.text(0.05,0.05, 'O(0,0)')
plt.plot(A[0],A[1],'ro'); plt.text(A[0]+0.1,A[1]+0.1, 'A(2,0)')
plt.plot(B[0],B[1],'ro'); plt.text(B[0]-0.4,B[1]+0.1, 'B(-1,3)')
```

```
plt.plot(C[0],C[1], 'ro'); plt.text(C[0]-0.4,C[1]-0.2, 'C(-1,-3)')
plt.plot(M[0],M[1], 'go'); plt.text(M[0]+0.1,M[1]+0.1, 'M')

plt.axis('equal')
plt.xlim(-3,3); plt.ylim(-3,3)
plt.grid(alpha=0.3)
plt.xlabel('x'); plt.ylabel('y')
plt.title('Circle + Equilateral Triangle (Median=3a)')
plt.legend()
plt.tight_layout()
plt.show()
```

Python Code

```
import numpy as np
import matplotlib.pyplot as plt

# Given data
median_length_coefficient = 3 # coefficient of 'a' in median
    length

# Matrix approach to find radius
theta = 2 * np.pi / 3
cos_theta = np.cos(theta)
cos_theta_minus_1 = cos_theta - 1 # -3/2
coefficient_squared = cos_theta_minus_1**2 # 9/4

# Solve for r^2 coefficient:  $r^2 * (9/4) = 9a^2 \Rightarrow r^2 = 4a^2$ 
r_squared_coefficient = (median_length_coefficient**2) /
    coefficient_squared # 4
r = 2 # since a = 1 for plotting
```

Python Code

```
# Print final answer
print(f"Answer: (c)  $x + y = \{\text{int}(r\_squared\_coefficient)\}a$ ")
    return foot

# Define triangle vertices on the circle
A = np.array([r, 0])
B = np.array([r * cos_theta, r * np.sin(theta)])
C = np.array([r * cos_theta, -r * np.sin(theta)])
midpoint_BC = (B + C) / 2

# Plot setup
plt.figure(figsize=(8, 8))

# Draw circle
theta_circle = np.linspace(0, 2*np.pi, 200)
x_circle = r * np.cos(theta_circle)
y_circle = r * np.sin(theta_circle)
plt.plot(x_circle, y_circle, 'b-', linewidth=2, label=f'Circle:  $x + y = \{\text{int}(r\_squared\_coefficient)\}a$ ')
```

```
# Draw circle
theta_circle = np.linspace(0, 2*np.pi, 200)
x_circle = r * np.cos(theta_circle)
y_circle = r * np.sin(theta_circle)
plt.plot(x_circle, y_circle, 'b-', linewidth=2, label=f'Circle: x
        + y = {int(r_squared_coefficient)}a')

# Draw triangle
triangle_x = [A[0], B[0], C[0], A[0]]
triangle_y = [A[1], B[1], C[1], A[1]]
plt.plot(triangle_x, triangle_y, 'r-', linewidth=2, label='
        Equilateral Triangle')

# Draw median from A to midpoint of BC
plt.plot([A[0], midpoint_BC[0]], [A[1], midpoint_BC[1]], 'g--',
        linewidth=2, label='Median = 3a')
```



```
# Mark and label points
```

```
plt.plot(0, 0, 'ko') ; plt.text(0.05, 0.05, 'O(0,0)', fontsize  
=12)
```

```
plt.plot(A[0], A[1], 'ro') ; plt.text(A[0]+0.1, A[1]+0.1, 'A(2,0)  
, fontsize=12)
```

```
plt.plot(B[0], B[1], 'ro') ; plt.text(B[0]-0.4, B[1]+0.1, 'B(-1,3  
)', fontsize=12)
```

```
plt.plot(C[0], C[1], 'ro') ; plt.text(C[0]-0.4, C[1]-0.2, 'C(-1,-  
3)', fontsize=12)
```

```
plt.plot(midpoint_BC[0], midpoint_BC[1], 'go') ; plt.text(  
midpoint_BC[0]+0.1, midpoint_BC[1]+0.1, 'M', fontsize=12)
```

```
# Formatting
plt.axis('equal')
plt.xlim(-3, 3)
plt.ylim(-3, 3)
plt.grid(True, alpha=0.3)
plt.xlabel('x')
plt.ylabel('y')
plt.title('Circle with Equilateral Triangle (Median = 3a)')
plt.legend()

plt.tight_layout()
plt.show()
```